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UNIX PATHNAME RESOLUTION VISUALIZER

INTERACTIVE SIMULATION OF UNIX NAMEI() PATH RESOLUTION

CSA0403-OPERATING SYSTEM

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INTRODUCTION

- UNIX/Linux uses pathname resolution to locate files and directories in the filesystem.
- A pathname such as `/home/student/Documents/file.txt` is processed component by component.
- The `namei()` utility helps trace how each component of a pathname is resolved.
- During resolution, the operating system handles directories, inodes, symbolic links, . and ...
- However, this process is mostly hidden from users and can be difficult for students to visualize.
- Our project, UNIX Pathname Resolution Visualizer, provides an interactive way to understand and trace this process.



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OBJECTIVES

- Understand the UNIX pathname resolution process.
- Simulate component-by-component lookup.
- Demonstrate the role of directories and inodes.
- Handle special pathname components such as . and ...
- Detect symbolic-link traversal and resolution failures.
- Provide an interactive operating-systems learning tool.



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PROBLEM STATEMENT

- UNIX pathname resolution happens internally inside the operating system.
- Users cannot easily see how each pathname component is processed.
- Directories, inodes, symbolic links, . and .. are difficult to understand only through theory.
- Existing command-line tools provide information but not an interactive explanation.
- Goal: visually demonstrate the hidden process from a pathname to its final filesystem object.



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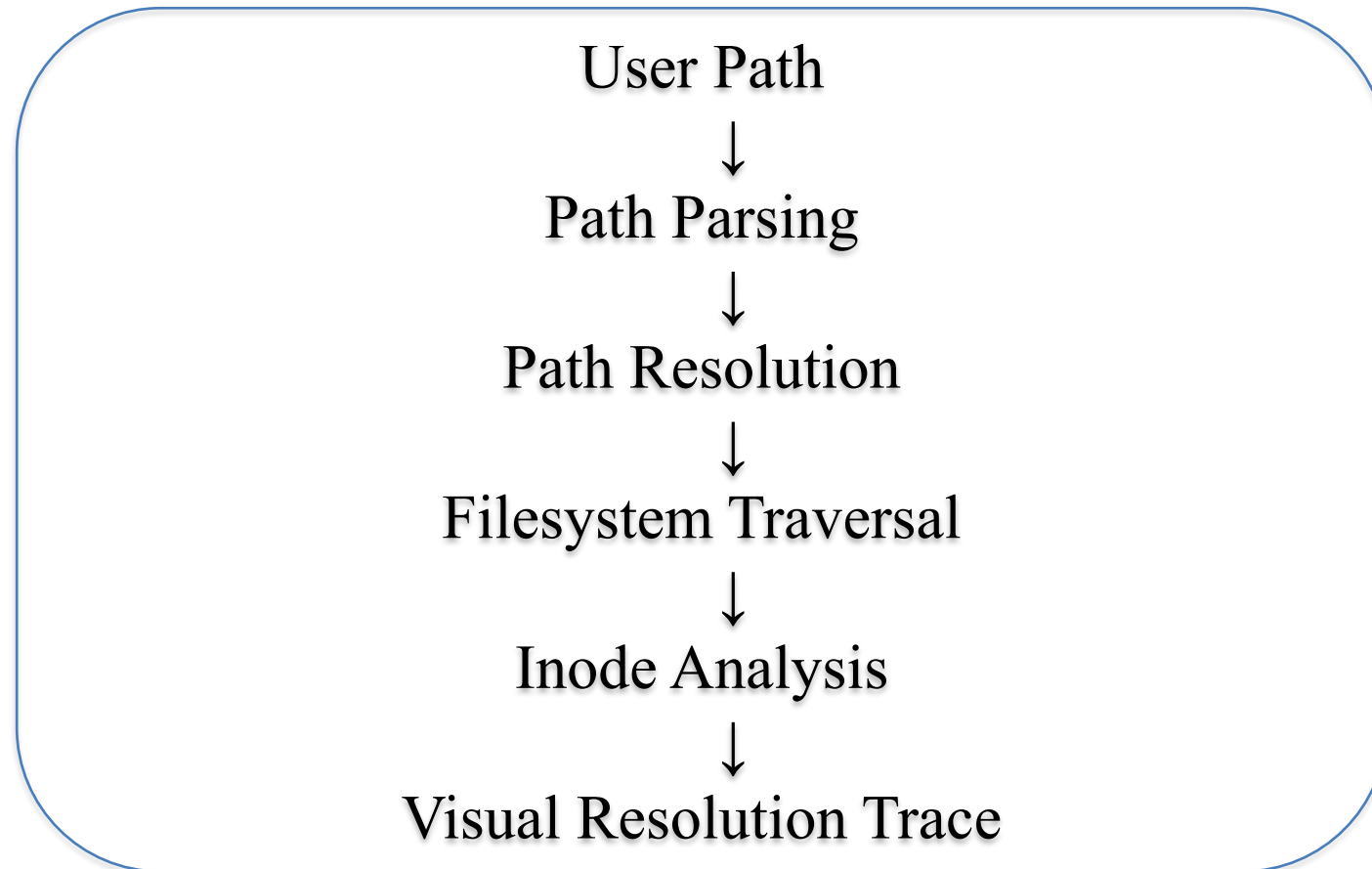


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EXISTING SYSTEM

- Common UNIX/Linux tools: namei, ls, stat and cd.
- Limitations: command-line oriented and difficult for beginners.
- Resolution steps are not presented as an interactive visual flow.
- Error location is not clearly highlighted.
- No integrated learning-oriented visualization.

PROPOSED SYSTEM



MODULE 1 — PATHNAME PARSING & VALIDATION

- Accepts a UNIX-style pathname.
- Identifies absolute or relative paths.
- Splits the pathname into individual components.
- Detects special components: . and ...
- Validates the input before resolution.

/home/student/Documents/project.txt



/ | home | student | Documents | project.txt

MODULE 2 — PATH RESOLUTION & FILESYSTEM TRAVERSAL

- Core module of the project.
- Resolves every component sequentially.
- Checks existence and whether a component is a directory.
- Handles . , .. and symbolic links.
- Stops at the exact component where resolution fails.

/ → home ✓ → student ✓ → Documents ✓ → project.txt ✓

MODULE 3 — INODE ANALYSIS & VISUALIZATION

- Retrieves metadata for the resolved filesystem object.
- Displays name, type, inode number, size and permissions.
- Shows the complete component-by-component resolution trace.
- Connects pathname lookup with the inode-based UNIX filesystem model.

Name: project.txt Type: Regular File
Inode: 184726 Permissions: rw-r--r--
Size: 52 bytes



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TECHNOLOGY STACK

- Frontend: HTML, CSS, JavaScript
- Backend: Python + Flask
- Filesystem operations: os, pathlib, stat
- Environment: Laptop/PC with browser
- Safe sandbox filesystem for controlled demonstration
- Upload component: browser file picker → sandbox/uploads/



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RESULT

The developed **UNIX Pathname Resolution Visualizer** successfully demonstrates the pathname resolution process through an interactive interface.

- Successfully parses and validates UNIX pathnames.
- Resolves pathname components step-by-step.
- Handles special components such as . and ...
- Detects and follows symbolic links.
- Displays the final file/directory and its inode metadata.
- Identifies errors such as ENOENT, ENOTDIR, EACCES, and ELOOP.
- Supports manual pathname entry and local file upload.



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FUTURE SCOPE

- Compare output directly with Linux namei -l
- System-call tracing using strace
- Interactive filesystem tree
- Step-by-step animation controls
- Cache vs non-cache lookup comparison
- Advanced permission simulation
- Multi-user filesystem simulation
- Path-resolution performance benchmarking



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CONCLUSION

- UNIX pathname resolution is an important filesystem operation.
- Our application makes this hidden process visible and interactive.
- The three modules parse, resolve and analyze a pathname.
- The system demonstrates directories, . and .., symbolic links and inodes.
- Errors are identified at the exact stage where resolution fails.
- Overall, the project transforms pathname resolution into an easy-to-understand visual experience.



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THANKYOU

